

SABR Volatility Curves, BAW Option Pricing and Historical-Simulation VaR with Full Revaluation

**An implementation and evaluation of the SR-FICC-2013-02
margining methodology on SPY options, with cross-vendor validation
and a two-regime comparison**

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Abstract

We implement and evaluate the options margining methodology described in a 2013 Fixed Income Clearing Corporation rule filing with the U.S. Securities and Exchange Commission (Release No. 34-69470, File No. SR-FICC-2013-02), which specifies three interlocking components: a SABR stochastic-volatility curve for the implied volatility smile, the Barone-Adesi and Whaley (1987) quadratic approximation for American option prices, and a 99% historical-simulation Value-at-Risk in which every scenario is *fully revalued* rather than approximated by a delta/gamma expansion. The filing governs options on interest rate futures; we apply the same machinery to SPY equity index options, which forces the general cost-of-carry form of the pricer that the filing’s futures options ($b = 0$) never exercise. The methodology is run end to end on two independent reference cases drawn from two independently sourced datasets: 2023-12-29 (OptionsDX end-of-day chains) and 2026-06-01 (Databento OPRA quotes, 353 sessions, acquired for \$8.11). The two cases give 99% one-day VaR of \$2.4488 and \$4.1216, or 36.89% and 28.58% of position value. To establish that this difference is a property of the market rather than of the data feed, we price the *original* reference date from the second vendor using the unmodified pipeline: the median mid-price difference across 151 common strikes is \$0.0000 and the median re-implied volatility difference is 0.0000 vol pts, with the put-call-parity forward agreeing to 0.89 basis points. That check also resolves a diagnostic left open by the implementation, showing that a wrong-signed kink of +0.291 vol pts at the put/call crossover was a vendor artefact while a residual inconsistency of roughly +0.15 vol pts beneath it is a genuine market feature reproduced by both sources to 0.0007 vol pts. We quantify two precision floors that the methodology cannot escape: the quadratic approximation contributes an error of approximately 1% to every scenario P&L, amplified 5.5× relative to the price-level error because its sign changes with moneyness; and the filing’s phrase “linear interpolation” does not determine a quantile rank convention, an ambiguity worth 1.46% of VaR in the first case but 21.26% in the second. The dominant sensitivity in both cases is neither of these but the composition of the lookback window, which we demonstrate within a single continuous dataset by watching the April 2025 selloff age out of the window between March and June 2026.

1 Introduction

In April 2013 the Fixed Income Clearing Corporation filed a proposed rule change with the Securities and Exchange Commission seeking to admit options on short-dated interest rate futures into

the one-pot cross-margining program it operated jointly with New York Portfolio Clearing (FICC, 2013). Admitting options to a margining program that had previously handled only linear instruments required the filing to say how those options would be valued and how their risk would be measured. The answer it gives rests on three components, and the interest of the specification lies in how tightly they are coupled.

A SABR curve (Hagan et al., 2002) supplies an implied volatility at any strike, which is what makes it possible to price a contract whose own quote is stale or absent. The Barone-Adesi and Whaley (1987) quadratic approximation converts that volatility into an American option price, which is required because the contracts carry an early-exercise right that a European formula prices at zero. Historical simulation generates roughly a year of alternative next-day market states by replaying observed daily moves. The binding constraint — and the reason the first two components are needed at all — is that each of those states is *fully revalued*: a fresh volatility from the curve and a fresh price from the pricer, never a Taylor expansion in the Greeks. A margining methodology that permitted the expansion would need neither a smile model nor an American pricer inside its scenario loop, since delta and gamma computed once at the reference state would suffice.

1.1 Why SPY rather than interest rate futures options

We apply the methodology to options on SPDR S&P 500 ETF Trust (SPY) shares rather than to the filing’s interest rate futures options. Three reasons, in decreasing order of importance.

First, data. Options on two-year Treasury futures are not available to us at the strike-by-strike, day-by-day granularity that a smile calibration and a 250-day historical simulation both require; SPY chains are, from more than one vendor, which is what makes the cross-vendor validation in Section 3.3 possible at all. Second, the substitution *exercises more of the methodology, not less*. Options on futures carry a cost of carry $b = 0$, so the pricer collapses to the Black (1976) form and the American premium on a call is structurally different. SPY is a dividend-paying equity ETF with $b = r - q$, so the general cost-of-carry pricer must be implemented and b must be obtained from somewhere. We derive it from the quotes rather than assume it, and one of the two reference cases turns out to sit on each side of the $b = r$ boundary that determines whether an American call has any early-exercise value at all. Third, SPY options are American-style, as the filing’s are, so the choice of pricer is not a matter of convenience.

What is lost is that SPY is not the filing’s underlying and its smile is not the filing’s smile. This is an evaluation of a *method*, and no claim is made that the numbers here would be the numbers a clearing house computes.

1.2 Contributions

Beyond implementing the specified methodology, this paper contributes three things.

An independent-vendor validation of the reference market state. Every result in a study of this kind rests on a single day’s quotes from a single vendor, and that dependence is usually acknowledged and then left alone. We instead purchased the same reference date from a second, structurally different source — consolidated OPRA quotes rather than an end-of-day vendor file — and ran the project’s own unmodified parity and inversion code against it. Section 3.3 reports what agreed, what did not, and, importantly, one diagnostic of ours that the second source showed to be unreliable.

A second complete reference case, in a different regime and from a different source.

The same code, with only a configuration file and a data directory changed, is run on a 2026 reference date built from a dataset we assembled ourselves. Because the cross-vendor check bounds the feed’s contribution, the differences between the two answers can be attributed to the market with a quantitative bound rather than an assertion.

A within-dataset demonstration of lookback-window aging. The standard criticism of unweighted historical simulation — that a crisis counts fully until the day it leaves the window and not at all thereafter — is usually illustrated by comparing different eras. We show it happening inside one continuous seventeen-month dataset, as the April 2025 selloff falls off the back of a rolling 250-day window.

We also report several negative and null results plainly, because they bound what this evaluation establishes: a residual put/call volatility inconsistency that re-impling the volatilities reduced but did not remove; a diagnostic metric of our own that a second data source falsified; and the fact that the second reference case, unlike the first, has no independent price benchmark of its own.

2 Methodology

Throughout, S is the spot price of the underlying, K a strike, T the time to expiry in years on an ACT/365 basis, r the continuously compounded risk-free rate, q a continuous dividend yield, $b = r - q$ the cost of carry, and $F = Se^{bT}$ the forward. Volatilities are quoted in decimal form but differences between them are reported in *volatility points*, one point being 0.01 in decimal terms.

2.1 The market state: forward, carry and the out-of-the-money smile

The smile model is written in terms of the forward, so the forward must be established before anything is calibrated. Rather than assume a dividend yield, we pin r externally to the three-month Treasury constant-maturity yield (Federal Reserve Bank of St. Louis, 2026) and imply the forward from put-call parity across liquid near-the-money strikes,

$$F = K + e^{rT}[C(K) - P(K)], \quad (1)$$

averaging the per-strike estimates, and then recovering the carry and the implied dividend yield by inversion,

$$b = \frac{1}{T} \ln \frac{F}{S}, \quad q = r - b. \quad (2)$$

The dispersion of the per-strike estimates entering (1) is the diagnostic that the procedure worked, and is reported for both cases in Section 4.

Equation (1) holds exactly only for European options, and SPY options are American, so the put’s early-exercise premium biases F slightly low. We quantify that bias rather than ignore it (Section 5.1).

The smile itself is built from out-of-the-money quotes only, split at the forward rather than at spot: below F the put is taken, above F the call. The in-the-money leg is illiquid and its quoted volatility unreliable — at one strike in the 2023 chain the deep in-the-money call implies 27.2% against 23.2% for the liquid out-of-the-money put at the same strike — and including it would give the fit two mutually inconsistent observations per strike.

2.2 SABR

Under SABR (Hagan et al., 2002) the forward and its volatility follow

$$dF_t = \alpha_t F_t^\beta dW_t^1, \quad d\alpha_t = \nu \alpha_t dW_t^2, \quad dW_t^1 dW_t^2 = \rho dt, \quad (3)$$

with $\alpha_0 = \alpha > 0$ the volatility level, $\beta \in [0, 1]$ the elasticity, ρ the spot/volatility correlation and $\nu \geq 0$ the volatility of volatility. The model is used through Hagan et al.'s singular perturbation expansion for the Black implied volatility, which for $K \neq F$ reads

$$\sigma_B(K, F) = \frac{\alpha}{(FK)^{(1-\beta)/2} \left[1 + \frac{(1-\beta)^2}{24} \ln^2 \frac{F}{K} + \frac{(1-\beta)^4}{1920} \ln^4 \frac{F}{K} \right]} \cdot \frac{z}{x(z)} \cdot [1 + \Theta T], \quad (4)$$

where

$$z = \frac{\nu}{\alpha} (FK)^{(1-\beta)/2} \ln \frac{F}{K}, \quad x(z) = \ln \left[\frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho} \right], \quad (5)$$

$$\Theta = \frac{(1-\beta)^2 \alpha^2}{24 (FK)^{1-\beta}} + \frac{\rho \beta \nu \alpha}{4 (FK)^{(1-\beta)/2}} + \frac{(2-3\rho^2) \nu^2}{24}. \quad (6)$$

The factor $z/x(z)$ in (4) is 0/0 at the money. We do not regularise it numerically; we branch explicitly on $|\ln(F/K)|$ below a tolerance to the closed-form limit $z/x(z) \rightarrow 1$, giving

$$\sigma_B(F, F) = \frac{\alpha}{F^{1-\beta}} \left[1 + \left(\frac{(1-\beta)^2 \alpha^2}{24 F^{2-2\beta}} + \frac{\rho \beta \nu \alpha}{4 F^{1-\beta}} + \frac{(2-3\rho^2) \nu^2}{24} \right) T \right]. \quad (7)$$

The branch was verified continuous: probing either side of the tolerance and subtracting the genuine slope contribution implied by the model's own skew leaves an unexplained excess of 4.5×10^{-8} volatility points, which is floating-point noise.

Calibration. With β fixed, the remaining three parameters are fitted by unweighted least squares to the observed out-of-the-money volatilities $\hat{\sigma}_i$ over a restricted set of strikes,

$$(\hat{\alpha}, \hat{\rho}, \hat{\nu}) = \arg \min_{\alpha, \rho, \nu} \sum_{i \in \mathcal{M}} [\sigma_B(K_i, F; \alpha, \beta, \rho, \nu, T) - \hat{\sigma}_i]^2, \quad \mathcal{M} = \left\{ i : \left| \frac{K_i}{F} - 1 \right| \leq 0.15 \right\}. \quad (8)$$

The optimisation is run from twelve starting points with bounds that scale correctly with β ; the upper bound on α is $5F^{1-\beta}$, since at $\beta = 1$ the parameter is a proportional volatility of order 0.1 while at $\beta = 0$ the model is normal and α is an absolute volatility in dollars of order 50.

Why β is fixed and not fitted. The filing specifies SABR but states no value for β . We fix $\beta = 1$. The reason is not convention but identification: β and ρ both control the skew and trade off against one another almost exactly, so a joint fit wanders along a flat valley of the objective. Table 3 makes this quantitative — across the full range $\beta \in [0, 1]$ the root-mean-square fit error moves by 0.04 vol pts, which is nothing, while $\hat{\alpha}$ swings by a factor of 476 and $\hat{\rho}$ slides to compensate. Fitting both would produce parameters that jump from day to day while the fit quality never improves. Setting $\beta = 1$ makes the model lognormal, matching the convention in which equity index volatilities are quoted, and leaves ρ as the sole skew parameter.

A second consideration is cross-asset, and it runs the other way. A native fixed-income application of this methodology would point towards *Normal* SABR, $\beta = 0$: on options on interest rate futures the forward can sit near zero or turn negative, where lognormal dynamics are undefined, and the additive dynamics of (3) at $\beta = 0$ remain well posed and match the convention in which rate volatility is quoted. Fixing $\beta = 1$ here follows from the move to an equity index underlying, whose forward is strictly positive and whose volatilities are quoted lognormally. The elasticity is thus the one structural parameter that the change of asset class genuinely determines; carrying the remaining components across unchanged is itself a test of how far the specification generalises beyond the instrument it was written for.

Why the calibration band is restricted. Equation (4) is an asymptotic expansion whose accuracy degrades away from the money, and the far wings of an SPY chain are penny quotes: in the 2023 chain, 35 of the 151 out-of-the-money strikes have a mid below \$0.10 with a median relative spread of 40%. In an unweighted objective such as (8) those quotes carry the same weight as at-the-money quotes with a one-cent spread, and they would drag the fit away from the region that matters. Restricting $\mathcal{M}$ to $|K/F - 1| \leq 0.15$ addresses both problems at once; Table 3 reports the sensitivity to that choice.

2.3 Barone-Adesi and Whaley

The generalised Black-Scholes-Merton price with cost of carry (Black and Scholes, 1973; Merton, 1973) is

$$c(S, K, T) = Se^{(b-r)T}N(d_1) - Ke^{-rT}N(d_2), \quad d_1 = \frac{\ln(S/K) + (b + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (9)$$

which reduces to Black's futures-option formula at $b = 0$ and to Black-Scholes at $b = r$. Barone-Adesi and Whaley (1987) approximate the American price as the European price plus a premium of closed form. Writing

$$M = \frac{2r}{\sigma^2}, \quad \mathcal{N} = \frac{2b}{\sigma^2}, \quad k = 1 - e^{-rT}, \quad q_{1,2} = \frac{-(\mathcal{N} - 1) \mp \sqrt{(\mathcal{N} - 1)^2 + 4M/k}}{2}, \quad (10)$$

the American call and put are

$$C(S, K, T) = \begin{cases} c(S, K, T) + A_2 \left(\frac{S}{S^*}\right)^{q_2}, & S < S^*, \\ S - K, & S \geq S^*, \end{cases} \quad A_2 = \frac{S^*}{q_2} \left[1 - e^{(b-r)T}N(d_1(S^*))\right], \quad (11)$$

$$P(S, K, T) = \begin{cases} p(S, K, T) + A_1 \left(\frac{S}{S^{**}}\right)^{q_1}, & S > S^{**}, \\ K - S, & S \leq S^{**}, \end{cases} \quad A_1 = -\frac{S^{**}}{q_1} \left[1 - e^{(b-r)T}N(-d_1(S^{**}))\right], \quad (12)$$

where the critical prices S^* and S^{**} — the exercise boundaries — solve the value-matching conditions

$$\begin{aligned} S^* - K &= c(S^*, K, T) + \left[1 - e^{(b-r)T}N(d_1(S^*))\right] \frac{S^*}{q_2}, \\ K - S^{**} &= p(S^{**}, K, T) - \left[1 - e^{(b-r)T}N(-d_1(S^{**}))\right] \frac{S^{**}}{q_1}. \end{aligned} \quad (13)$$

We solve (13) by Newton-Raphson from the standard seed, with each iterate clamped into the region where a boundary can economically lie ($S^* > K$ for a call, $S^{**} \in (0, K)$ for a put) and with an explicit error raised for a call when $b \geq r$, where no finite boundary exists.

A European price will not serve here. When $b < r$ the American call carries a genuine early-exercise premium, and the American put carries one in every regime; on the 2023 reference contract that premium is \$0.29 on a \$6.47 price, or 4.7% of the option's value, which is not a rounding error in a margin calculation. The approximation itself is not exact, and Section 4.1.4 measures what it costs.

2.4 Historical simulation with full revaluation

Let S_0 be the reference spot and $\{S_{-N}, \dots, S_0\}$ the closing price series over the lookback. Simple daily returns

$$r_i = \frac{S_{-N+i} - S_{-N+i-1}}{S_{-N+i-1}}, \quad i = 1, \dots, N, \quad (14)$$

are applied to the reference spot to generate $N = 250$ alternative next-day states. For each,

$$\tilde{S}_i = S_0(1 + r_i), \quad \tilde{F}_i = \tilde{S}_i e^{bT}, \quad \tilde{\sigma}_i = \sigma_B(K, \tilde{F}_i; \hat{\alpha}, \hat{\beta}, \hat{\rho}, \hat{v}, T), \quad \tilde{V}_i = V^{\text{BAW}}(\tilde{S}_i, K, T, r, b, \tilde{\sigma}_i), \quad (15)$$

and the scenario profit and loss for a long position of one contract is $L_i = \tilde{V}_i - V_0$, where V_0 is the reference price computed the same way.

Three properties of (15) are decisions rather than mechanics. The calibrated parameters are held fixed and the curve is re-evaluated at the *new* forward, so the smile is pinned to moneyness rather than to strike. Time to expiry is held fixed at its reference value, excluding theta, which would otherwise shift every scenario down by approximately the same amount and inflate the VaR irrespective of market direction. And the volatility surface is not itself shocked: $\tilde{\sigma}_i$ moves only because $\tilde{F}_i$ moves. The consequences of all three are quantified in Section 6.

The step the filing forbids is the Taylor alternative,

$$L_i^\Delta = \Delta(\tilde{S}_i - S_0), \quad L_i^{\Delta\Gamma} = \Delta(\tilde{S}_i - S_0) + \frac{1}{2}\Gamma(\tilde{S}_i - S_0)^2. \quad (16)$$

We compute (16) deliberately, as a counterfactual only, to measure in dollars what the prohibition buys; it is never used for a reported risk number.

2.5 Value-at-Risk by interpolation between order statistics

Let $L_{(1)} \leq L_{(2)} \leq \dots \leq L_{(N)}$ be the ordered scenario P&Ls and let $p = 1 - c = 0.01$ be the tail probability at confidence $c = 99\%$. The filing specifies linear interpolation to the threshold. Writing the fractional rank as

$$h = (N - 1)p, \quad k = \lfloor h \rfloor, \quad w = h - k, \quad (17)$$

the interpolated quantile and the reported risk measures are

$$\hat{Q} = L_{(k+1)} + w(L_{(k+2)} - L_{(k+1)}), \quad \text{VaR} = -\hat{Q}, \quad \text{ES} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} L_i, \quad \mathcal{B} = \{i : L_i \leq \hat{Q}\}. \quad (18)$$

VaR and expected shortfall are reported throughout as *positive loss magnitudes*; the quantile retains its natural sign and is negated once, at the point of reporting. A position that lost nothing even at

its first percentile would produce a negative VaR, and our implementation flags that case rather than clipping it to zero.

Equation (17) is a choice, and the filing’s wording does not make it. The definition $h = (N - 1)p$ is R’s type 7 and NumPy’s default; $h = Np$ and $h = (N + 1)p - 1$ (R’s type 6, the Weibull convention) are both in common use (Hyndman and Fan, 1996). We implement the interpolation arithmetically rather than by library call, precisely so that the convention is a stated assumption rather than an inherited default, and we report the sensitivity in Section 5.2. Expected shortfall as defined in (18) is the simple mean of the breaching scenarios rather than an integrated tail expectation (Acerbi and Tasche, 2002); at $N = 250$ and $p = 0.01$ only three scenarios breach, so it is correspondingly noisy.

Weighting. The simulation is unweighted: every day in the window counts equally. This matches the filing and departs from common practice, which would apply exponential decay (Boudoukh et al., 1998) or rescale historical returns by the ratio of current to historical volatility (Barone-Adesi et al., 1999). The deviation is deliberate, made to match the specification rather than to improve the estimate, and Section 4.4 shows what it costs.

3 Data

Two datasets support two reference cases. They differ in vendor, in structure, in the period they cover and in how much of the preparation we did ourselves, which is the point: a methodology that transfers between them has been shown to transfer, not asserted to.

3.1 OptionsDX end-of-day chains, 2010–2023

The first case uses SPY end-of-day option chains published by OptionsDX (OptionsDX, 2024) and widely mirrored, fourteen annual files covering 2010–2023. Each row is one (quote date, expiry, strike) triple carrying both the call and the put leg with bid, ask, last, volume and a vendor-supplied implied volatility. Raw column names are wrapped in square brackets, a vendor quirk stripped on load.

The underlying price series is taken from the `UNDERLYING_LAST` field inside these same files rather than from a separate equity feed, so that spot remains synchronous with the option quotes it is paired against. That field is the unadjusted SPY price, which means SPY’s four annual ex-dividend dates appear in the return sample as spurious down-moves of roughly 0.4%. We accept this: the option contracts are written on the unadjusted price, so shocking the unadjusted price is internally consistent, and none of the four falls inside the reference lookback window.

A vendor defect, found and bounded. Validating the price series against a derived NYSE calendar exposed a date-stamping fault in the 2022 file. Four genuine trading days are absent (2022-05-11, 06-17, 06-24 and 12-16) and nine rows are stamped on days the exchange was closed (including 2022-01-17, 07-04 and 11-24). The two faults damage a return sample in opposite directions. A missing session compresses two days of movement into one return and manufactures a tail observation that never occurred: the 2022-12-15 to 2022-12-19 interval appears as a single -2.49% move when it was in fact -1.65% followed by -0.85% . A phantom row on a closed market carries a stale price forward and injects a spurious near-zero return, diluting the tail instead. *Zero defects fall*

inside the 250-day window used for the reference case, which begins 2022-12-30; the check is re-run on every execution rather than assumed.

3.2 Databento OPRA quotes, 2025–2026

The second case uses a dataset we assembled from Databento (Databento, 2026): 353 trading sessions from 2025-01-02 to 2026-06-01, 11,036,871 quote records reshaped into 1,880,602 rows, acquired for \$8.11 in total.

Three schemas were used, each chosen for what it structurally contains. Consolidated best-bid/offer bars at one-minute resolution (`cbbo-1m`, `OPRA.PILLAR`) supply a two-sided quote for every listed strike including the illiquid wings, which is the whole smile. Trade-derived alternatives were rejected for exactly that reason: an option that does not trade produces no bar, and the far wings that pin the SABR skew and curvature trade rarely or never on a given day, so most of the smile would simply be absent. Instrument definitions (`definition`, same dataset) supply strike, expiry and option type for each numeric instrument identifier, without which a quote row cannot be attached to a strike at all. Daily consolidated bars for SPY itself (`ohlcv-1d`, `EQUUS.SUMMARY`) supply `UNDERLYING_LAST`, since OPRA carries no underlying price.

The pull takes a three-minute pre-close window, 15:57–16:00 ET, and keeps each contract’s last bar within it, so that a contract quoted only in an earlier minute keeps that quote rather than being dropped. The window is resolved per date with a timezone database rather than a hardcoded UTC offset: 16:00 ET is 20:00 UTC in summer and 21:00 UTC in winter, and getting that wrong silently returns the wrong three minutes.

Three preparation problems are worth recording because each was invisible to a naive approach and each is a class of defect rather than a one-off.

Early closes. The NYSE closes at 13:00 ET on roughly three days a year. Those remain trading days, so a derived calendar reports them as ordinary and a 16:00 window looks entirely normal while sitting three hours after the bell. Three such sessions in range returned no data at all; pulled at the 13:00 close they returned normal volumes of 26,022, 31,440 and 26,959 rows against a 30,273 median. This never surfaced in the 2010–2023 work because an end-of-day vendor stamps one row per day regardless of session length.

A one-off closure. 2025-01-09 failed differently, with no symbol resolving at all: the exchange was closed for a national day of mourning. The session count is therefore 353, not 354. The two failure modes are distinguishable — on a short session the symbols resolve and the window is merely empty — and the distinction is now documented in the acquisition code.

Definition coverage. Instrument definitions were first pulled quarterly, on the reasoning that they are largely static. They are not static enough: contracts listed between anchor dates have no definition, and the match rate decayed from 100% near an anchor to 89.3% by 2026-03-17. A per-session definition snapshot for every day in range took the match rate to 100.0000% across all 11,036,871 rows, at a cost of \$6.34 against \$0.12 — the reference data cost roughly four times the quote data.

One column in the reshaped dataset is not what its name suggests, and the consequence propagates into Section 4. The volatility columns `C_IV` and `P_IV` hold European Black-76 volatilities that we computed from the mid on the parity forward, not a vendor’s numbers. They exist because the smile constructor filters on that column and runs before the re-implying step can replace it, so a null column returns an empty smile. The inversion round-trips to 3.4×10^{-13} in price. The consequence is that the before/after comparison of Section 4.1.3 measures something different in

Table 1: Cross-vendor comparison of the 2023-12-29 reference state. The project’s own parity and inversion code was run unchanged against Databento OPRA quotes, so every difference is attributable to the data. Every quantity agrees within ordinary cross-vendor noise; the two entries that do not sit at zero are examined in the text.

Quantity	OptionsDX	Databento OPRA	Difference
Chain strikes, expiry 2024-02-16	156	156	0
Parity forward (\$)	478.8920	478.8495	−0.0425 (0.89 bp)
Forward dispersion (\$)	0.203 (22 strikes)	0.182 (32 strikes)	tighter
Implied dividend yield (%)	−0.1926	−0.1265	+0.066 pp
\$475 put, bid / ask (\$)	6.51 / 6.54	6.58 / 6.61	mid +0.0700
BAW theoretical – market mid (\$)	+0.1141	+0.0441	closer to theory
Crossed out-of-the-money quotes	2 ($K = 477, 478$)	0	—
<i>Across the 151 common out-of-the-money strikes</i>			
Median mid price difference (\$)	—	—	0.0000
Median re-implied IV difference	—	—	0.0000 vol pts
Mean re-implied IV difference	—	—	−0.0168 vol pts
95th percentile IV difference	—	—	0.2286 vol pts
Strikes differing by more than 1 vol pts	—	—	1 of 151
Re-implied crossover gap, 476 → 479	−0.2088	−0.2164	0.0076 vol pts

each case: in 2023 it measures vendor error *plus* the European-to-American correction, in 2026 the correction alone. The two are therefore not comparable baselines, and no inference about relative data quality should be drawn from the fact that the 2026 correction is the smaller of the two — it is smaller because it contains one term rather than two.

3.3 Cross-vendor validation of the reference market state

Before extending anything, we purchased the *original* reference date, 2023-12-29, from Databento and ran the project’s own `imply_forward_from_parity` and implied-volatility inversion against it without modification, so that every difference below is attributable to the data rather than to the method. Cost: \$0.0390.

The join was verified independently before any comparison: the OSI symbol carried on each quote matches the symbol on its definition for all 312 instruments in the reference expiry, with zero mismatches, and the joined chain yields 156 strikes with both legs — exactly the 156 the OptionsDX chain contains. Table 1 reports the comparison and Figure 1 shows it strike by strike.

What the check establishes. The OptionsDX-derived reference state is independently corroborated; the instrument mapping works; and the project’s parity and inversion code runs unchanged against a structurally different source and produces consistent results. This is what licenses treating any later difference between the two reference cases as a market difference rather than a feed difference, and Section 4.3 makes that argument quantitatively.

What it does not establish. It validates one date. A rebuild of a 2025–2026 period rests on a mapping confirmed on a single day, and no second source has been purchased for the 2026 reference date itself. Nor are the two objects being compared identical in kind: a consolidated best-bid/offer at a minute boundary is not the same thing as an end-of-day vendor snapshot. They agree

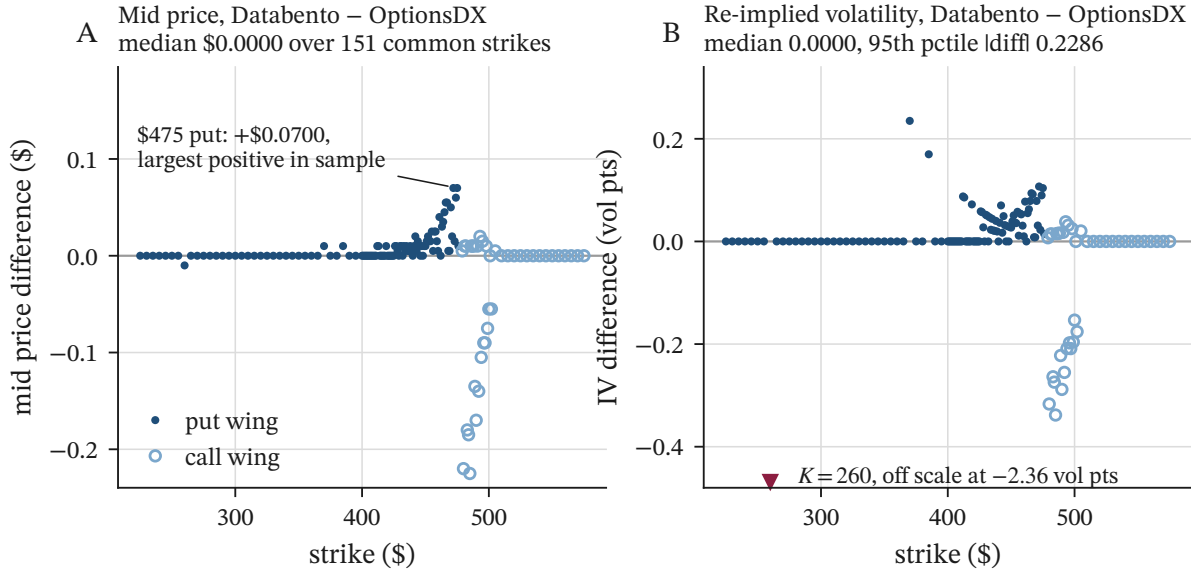


Figure 1: The 2023-12-29 reference date priced from two independent vendors, strike by strike. **A:** mid-price differences sit on zero across the chain; the median over 151 common strikes is exactly \$0.0000. The largest positive difference, +\$0.0700, falls on the \$475 put — the single contract the headline VaR rests on — and both vendors quote a three-cent spread there, so the gap is snapshot timing rather than disagreement about value. The negative cluster near \$490 is the call wing, where the two vendors’ snapshots diverge by up to $-\$0.225$. **B:** re-implied volatility differences, median 0.0000 vol pts. Exactly one strike of 151 differs by more than one volatility point and it is plotted off scale: $K = 260$ is a deep-wing penny put quoted 0.01/0.02 against a project mid of 0.025, so a one-cent disagreement is amplified into 2.36 vol pts by near-zero vega. Rescaling the axis to contain it would hide the finding, which is that the bulk lies within ± 0.35 vol pts.

closely here, but a rule for reducing minute bars to an end-of-day equivalent is a modelling decision, not a data pull, and it is ours. Finally, the “implied spot” agreement sometimes quoted in such comparisons is not available here and we do not report one: OPRA carries no underlying price, so spot cannot be read from these data at all.

4 Results

The methodology of Section 2 was run end to end twice. The pricing, calibration, simulation and VaR code is byte for byte identical between the two runs; only a configuration file and a data directory differ. Table 2 states both reference configurations, distinguishing quantities assumed from quantities derived from the quotes.

4.1 Case 1: 2023-12-29, OptionsDX

4.1.1 Market state and smile construction

The reference expiry carries 156 strikes from \$210 to \$575, reduced to a 151-strike out-of-the-money smile after quality filters, with quoted volatilities running from 9.94% to 64.20%. Twenty-two liquid near-the-money strikes enter the parity calculation (1) and agree on the forward to a standard deviation of \$0.203, roughly four basis points of spot; that tightness is the evidence the procedure

Table 2: The two reference configurations. Only two inputs are assumed — the risk-free rate, pinned externally to the same FRED series in both cases, and the SABR elasticity β . The forward, the cost of carry and the implied dividend yield are all derived from the quotes by (1) and (2); the SABR parameters are fitted by (8). The calibration band, the elasticity, the quote filters and the confidence level are deliberately held identical across the two cases, so that a comparison of results is not confounded by a comparison of method.

Quantity	2023-12-29	2026-06-01	Provenance
Data source	OptionsDX EOD	Databento OPRA	—
Reference expiry	2024-02-16	2026-07-17	standard monthly
Days to expiry	49	46	calendar, ACT/365
T (years)	0.13425	0.12603	derived
Spot S (\$)	475.31	758.54	observed close
Risk-free rate r (%)	5.4000	3.6600	assumed , FRED DGS3MO
Forward F (\$)	478.892	760.224	derived , put-call parity
Forward dispersion (\$)	0.203 (22 strikes)	—	derived
Cost of carry b (%)	5.5926	1.7601	derived
Dividend yield q (%)	−0.1926	+1.8999	derived
Carry regime	$b > r$	$b < r$	consequence
Chain strikes	156	—	observed
Smile strikes (OTM)	151 (112 P, 39 C)	197 (129 P, 68 C)	after quality filters
Calibration strikes	103	125	$ K/F - 1 \leq 0.15$
SABR β	1.0	1.0	assumed , not fitted
SABR $\hat{\alpha}$	0.109797	0.136311	fitted
SABR $\hat{\rho}$	−0.524225	−0.633712	fitted
SABR $\hat{\nu}$	2.132810	2.201313	fitted
Fit RMSE	0.2346 vol pts	0.2080 vol pts	fitted
Position	+1 × \$475 put	+1 × \$759 put	—
Moneyness of position	0.065% OTM	0.061% ITM	—
SABR volatility at K	11.7554%	13.9399%	derived
Position value V_0 (\$)	6.639087	14.422074	derived
Lookback window	2023-01-03 – 2023-12-29	2025-06-03 – 2026-06-01	250 sessions
Confidence level	99%	99%	from the filing

worked.

The forward implies a dividend yield of −0.1926% — essentially zero, and correctly so. SPY’s December 2023 ex-dividend date was the 15th and the next fell on 2024-03-15, so no dividend falls between the reference date and the expiry. Substituting SPY’s approximately 1.4% trailing yield would have mispriced the forward by about \$0.90, roughly four strike increments, and tilted the entire smile. The near-zero implied yield is independent corroboration that the 5.40% rate assumption is sound.

4.1.2 SABR calibration

Fitting (8) over the 103 strikes within 15% of the forward gives $\hat{\alpha} = 0.109797$, $\hat{\rho} = -0.524225$, $\hat{\nu} = 2.132810$, with a root-mean-square error of 0.2346 vol pts and a maximum absolute error of 0.5075 vol pts. The correlation parameter is negative, as an equity index requires. Panel A of Figure 2 shows the fit and its residuals.

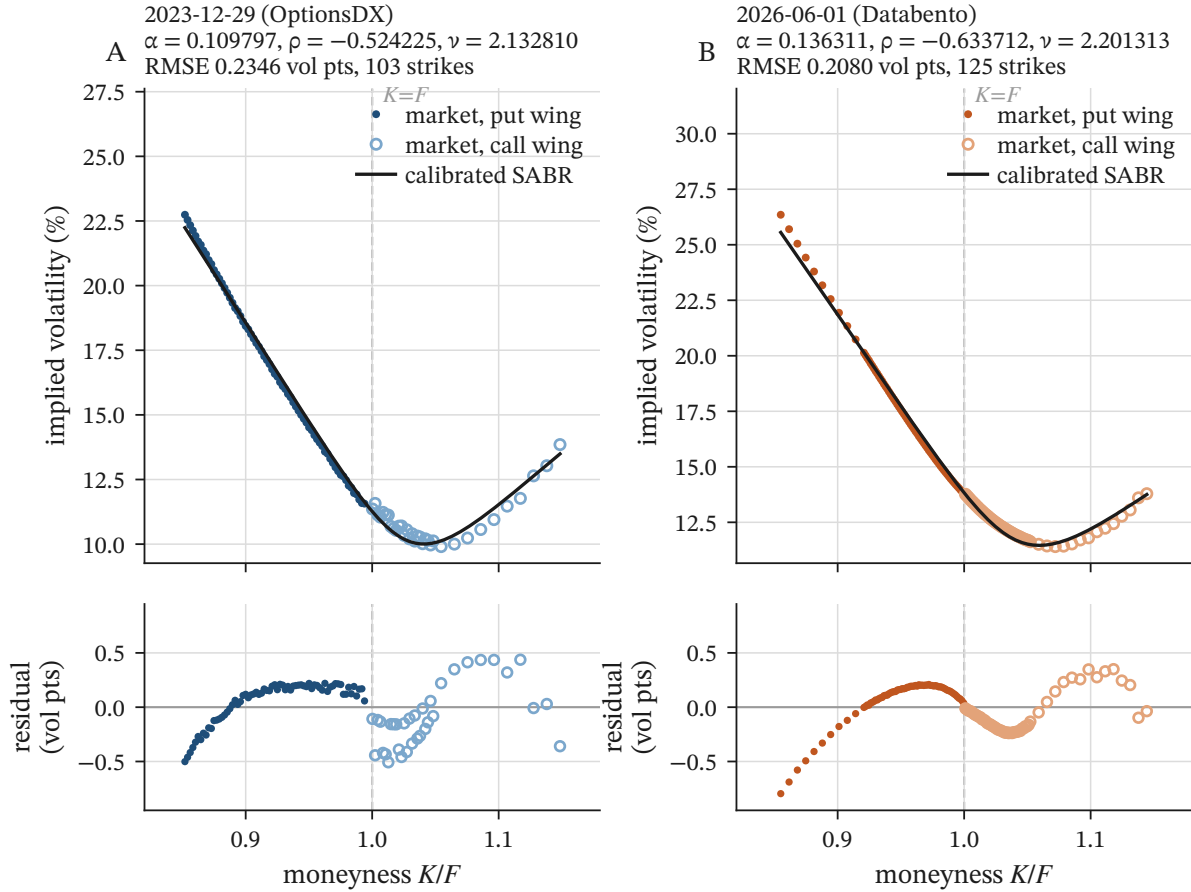


Figure 2: Calibrated SABR against the re-implied out-of-the-money smile, plotted against moneyiness K/F so that the two cases sit on a comparable axis. Solid markers are the put wing, open markers the call wing, and the black line is (4) evaluated at the fitted parameters. The 2026 skew is visibly steeper — $\hat{\rho}$ moves from -0.524 to -0.634 — which is legible only because the horizontal axis is shared; against strike the two panels span $\$408$ – $\$550$ and $\$650$ – $\$870$ and cannot be compared by eye. The residual panels show what the fit cannot reach. In both cases the residual changes sign at $K = F$, which is the signature of the two wings disagreeing with each other rather than of the model failing: fitted separately, each wing of the 2023 smile reaches its own quote-noise floor (Section 4.1.2).

The residual pattern in Figure 2 is a data property, not a model failure, and establishing that is what licenses the rest of the analysis. Fitting each wing separately and comparing the result to the strike-to-strike jitter in the quoted volatilities themselves — a true smile being smooth, that jitter is quote noise, and a model that reaches it has done as well as anything could — gives:

Fit	Strikes	RMSE (vol pts)	Quote-noise floor (vol pts)	$\hat{\rho}$
Both wings	103	0.2946	0.1140	-0.5108
Puts only	69	0.0751	0.0501	-0.2578
Calls only	34	0.2036	0.1851	-0.6008

(These figures are computed on the vendor’s volatilities, before the correction of Section 4.1.3, which is why the combined RMSE is 0.2946 rather than 0.2346.) Each wing individually fits to

Table 3: The identification problem, made quantitative. **Upper panel:** varying the elasticity β across its entire admissible range moves the fit error by 0.04 vol pts while $\hat{\alpha}$ swings by a factor of 476 and $\hat{\rho}$ slides to compensate — the objective has a flat valley and the data cannot separate β from ρ . This is why β is fixed rather than fitted. **Lower panel:** widening the calibration band degrades the fit smoothly, by 2.8 $\times$ at the full smile, while $\hat{\rho}$ stays within -0.48 to -0.52 throughout. The skew is well determined; it is the wings that Hagan’s asymptotic expansion cannot reach. Both panels are computed on the 2023-12-29 chain.

	Strikes	RMSE (vol pts)	$\hat{\alpha}$	$\hat{\rho}$	$\hat{v}$
<i>Elasticity β, calibration band fixed at 15%</i>					
$\beta = 0.00$	103	0.2548	52.3342	−0.4488	2.0076
$\beta = 0.30$	103	0.2653	8.2298	−0.4681	2.0352
$\beta = 0.50$	103	0.2730	2.3979	−0.4807	2.0542
$\beta = 0.70$	103	0.2813	0.6987	−0.4929	2.0737
$\beta = 1.00$ (used)	103	0.2946	0.1099	−0.5108	2.1038
<i>Calibration band $K/F - 1$, elasticity fixed at $\beta = 1$</i>					
5%	46	0.2040	0.1126	−0.5097	1.7349
10%	74	0.2349	0.1117	−0.5198	1.9425
15% (used)	103	0.2946	0.1099	−0.5108	2.1038
20%	118	0.3602	0.1086	−0.5033	2.1887
30%	128	0.5532	0.1057	−0.4985	2.3326
100%	151	0.8116	0.0991	−0.4784	2.5841

its own noise floor. The combined fit is worse than either because the two wings are mutually inconsistent, and the call wing is 3.7 $\times$ noisier than the put wing — SPY put quotes are simply better maintained at end of day.

Table 3 reports the two sensitivities that justify the fixed β and the restricted band.

4.1.3 Re-impling the volatilities

The vendor’s put and call implied volatilities are mutually inconsistent. Put-call parity requires a put and a call at the same strike to carry the same volatility; the vendor’s do not. We therefore discard them and recover every volatility ourselves, by inverting the same BAW pricer against the same parity forward that the rest of the analysis uses. All 151 out-of-the-money strikes inverted successfully.

The decisive evidence is the *sign* of the gap at the put/call crossover. The smile slopes downward through this strike region, so a self-consistent smile must show a negative step going from the last put to the first call. The vendor’s does not:

	Last put, $K = 476$	First call, $K = 479$	Adjacent gap
Vendor implied volatility	11.379%	11.670%	+0.291 vol pts (wrong sign)
Re-implied	11.567%	11.358%	−0.209 vol pts (correct sign)

No smooth curve can fit a kink pointing the wrong way, which is why the combined-wing fit of Section 4.1.2 could not reach either wing’s noise floor. Panel B of Figure 3 shows the reversal directly. After re-impling, the fit RMSE falls from 0.2946 to 0.2346 vol pts, the maximum error from 0.6378 to 0.5075 vol pts, and the put-wing quote roughness from 0.0501 to 0.0320 vol pts.

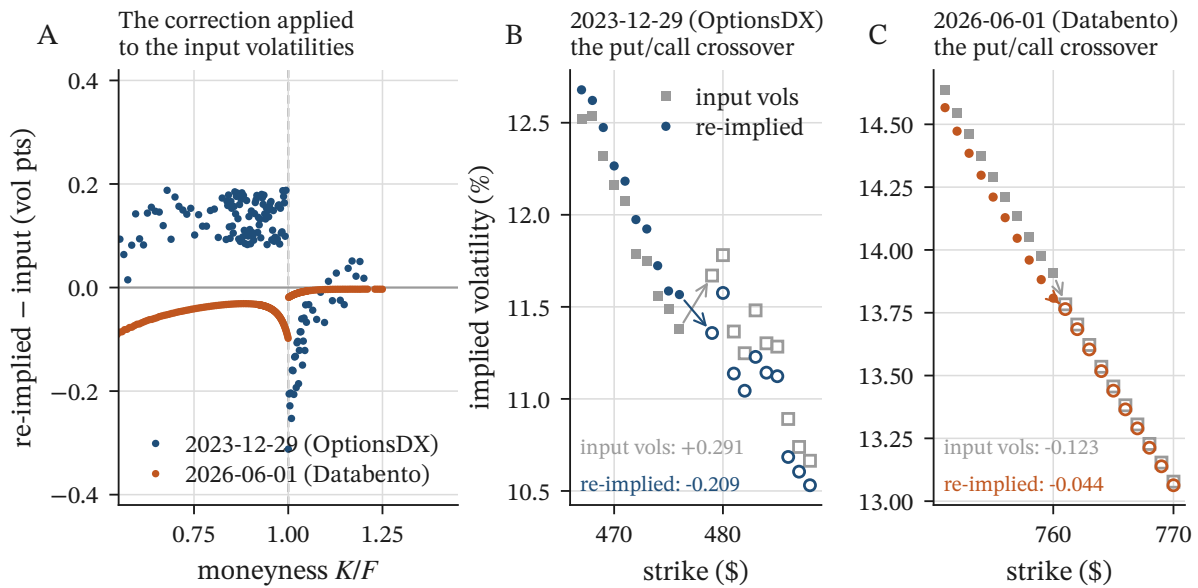


Figure 3: **A:** the correction applied to the input volatilities, by moneyness. In 2023 it is a systematic offset that flips sign at the forward — $+0.1231$ vol pts on out-of-the-money puts, -0.0911 vol pts on calls — which is the signature of a structural inconsistency, not of noise. In 2026 the correction is small and one-signed (-0.0388 vol pts across all strikes) because the input column there is this project’s own European inversion rather than a vendor’s, so the panel measures the European-to-American correction alone. **The two curves in panel A therefore do not share a baseline and should not be read against one another:** the 2023 curve is vendor quote error compounded with the European-to-American premium correction, while the 2026 curve isolates that premium correction exclusively. Their relative magnitudes say nothing about data quality. **B:** the 2023 put/call crossover. The grey arrow is the vendor’s step, $+0.291$ vol pts, pointing *upward* against a downward-sloping smile — a shape no smooth curve can fit. The blue arrow is the same step after re-impling, -0.209 vol pts, correctly signed. **C:** the same construction in 2026, where both the pre- and post-correction gaps already carry the correct negative sign. An internally consistent starting point does not produce the kink.

4.1.4 The BAW pricer and what the approximation costs

The pricer was built and validated with no involvement from the smile model: volatility is a plain input argument here. Table 4 lists the ten checks. Correctness rests on the exact structural properties — tests 1, 2, 3, 7 and 8, all of which hold to machine precision or better — and not on the approximation-quality comparisons, which measure the accuracy of the *method* rather than of the implementation.

Two properties of the approximation constrain everything downstream, and Figure 4 shows both.

Calls are exact in the 2023 regime. With $b = 5.5926\% > r = 5.4000\%$, early exercise on a call is never optimal, (13) has no finite solution, the pricer returns the European value and the lattice agrees to 0.02%. There is no premium to get wrong. This is precisely why the reference position is a put: a call would exercise none of the American machinery.

Puts carry a real error whose sign flips with moneyness. At the reference expiry, BAW recovers about 95% of the at-the-money put’s early-exercise premium ($\$0.2914$ against a true $\$0.3069$) while *overstating* it out of the money by more than a factor of two ($\$0.0362$ against $\$0.0167$). The absolute errors are small — $\$0.0155$ at the money, $\$0.0195$ at $K = 450$ — and the large relative error out of

Table 4: The BAW validation suite. Tests 1–3 and 7–8 are exact structural properties and establish implementation correctness; tests 4–6 measure the accuracy of the approximation itself and are reported rather than asserted. No benchmark table was transcribed from the original paper: reference values that could not be verified from the build environment would validate the pricer against fiction, so the reference is a convergent Cox-Ross-Rubinstein lattice (Cox et al., 1979), which is the method Barone-Adesi and Whaley used to assess their own approximation. Test 2 is the sharpest single check of a BAW implementation and it passes to *exactly* zero rather than to tolerance.

#	Check	Result
1	European put-call parity	worst error 2.8×10^{-14} across 6 cases
2	American call = European when $b = r$	worst spurious premium exactly 0.0, 5 cases
3	American $\geq$ European and $\geq$ intrinsic	0 violations in 1,350 grid points
4	BAW vs. lattice, stress grid	worst absolute error \$0.0599 (tolerance \$0.10)
5	BAW vs. lattice, SPY regime	worst absolute error \$0.1439 (tolerance \$0.20)
6	P&L error propagation	informational; see text
7	Monotonicity in S, σ, T	10 sequences, all correct
8	Critical exercise prices	correct side of strike, recede as σ rises
9	Implied-volatility round trip	128 cases, worst error 7.893×10^{-6}
10	SPY reference case	informational

the money is an artefact of a small denominator.

The consequence for the risk number is Panel B of Figure 4, and it is the opposite of what one might assume. It would be comfortable to expect the pricing error to cancel in the scenario-minus-base subtraction, since the same systematic bias sits in both terms. It does not. Because the error varies with spot and changes sign with moneyness, differencing two prices at two different spots leaves *more* error than either price carried alone: the worst P&L error of \$0.0852 is 5.5× the price-level error of \$0.0155. **Approximately 1% of every scenario P&L, and therefore of the final VaR, is an artefact of using BAW rather than a lattice.** The filing specifies BAW, so this is the intended behaviour rather than an error, but it is a real precision floor and it is not removable within the specification.

4.1.5 The SABR-to-BAW junction

The two models meet at a single point: the curve is read at the *forward* and the pricer is called from *spot*. Conflating the two is the classic error at this junction, and here they differ by \$3.58, more than three strike increments. Pricing all 103 calibration strikes through the joined models against the market gives a mean absolute error of \$0.0718 and a mean *signed* error of $-\$0.0007$: scatter without bias.

Only 6 of 103 theoretical prices land inside the quoted bid-ask spread, which looks alarming until the spreads are examined — the median SPY spread here is \$0.010, less than 1% of mid, so “inside the spread” demands matching the market to half a cent, a standard no smile model meets. Absolute error is the meaningful measure.

The test that matters is attribution. If the wiring were wrong — spot passed where the forward belongs, a mismatched day count, an incorrect carry — the resulting error would be structural and would not track the volatility fit error. Multiplying each strike’s volatility error by its vega and comparing predicts the observed pricing error almost exactly (Figure 5): 99.21% of the error is explained, with a correlation of 0.999974 and a mean unexplained residual of \$0.000567. Vega is computed here for attribution only and is never used to price a scenario.

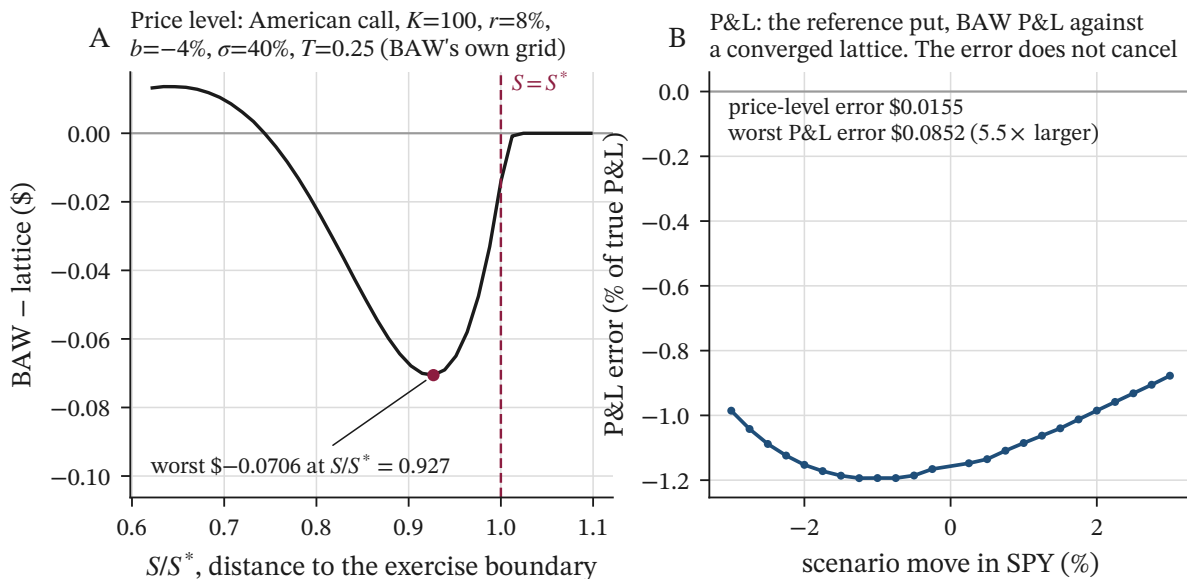


Figure 4: What the quadratic approximation costs. **A:** BAW minus a converged 8,000-step lattice for an American call on the parameter set Barone-Adesi and Whaley use for their own accuracy tables, plotted against distance to the exercise boundary S^* . The error is near zero at the money, peaks at $-\$0.0706$ at $S/S^* = 0.927$ — just below the boundary, where the term dropped from the pricing equation matters most — and is exactly zero beyond it, where the option is worth its intrinsic value. This is the approximation's documented weakness, not a defect: it is why the validation tolerances in Table 4 are set on absolute dollar error rather than relative error. (A five-point grid over the same parameters reports the peak as $\$0.0700$ at $S/S^* = 0.915$; the finer sampling here locates it slightly differently.) **B:** the consequence for a risk number. Scenario P&L is a *difference* of two prices at two different spots, and because BAW's error varies with spot and changes sign with moneyness, it does not cancel in that difference — it survives at approximately -1% of the true P&L across the entire range of daily moves a 250-day SPY sample contains. The worst P&L error, $\$0.0852$, is $5.5\times$ the $\$0.0155$ price-level error.

4.1.6 Scenarios and full revaluation

The 250 scenarios are drawn from 2023-01-03 to 2023-12-29 — exactly calendar 2023, since 250 sessions back from the reference date lands on the first trading day of the year — and produce scenario spot prices from $\$465.80$ to $\$486.16$. The window contains zero calendar defects. Its statistics appear in Table 7.

Three sanity checks on the revaluation loop pass. A scenario with a $+0.0000\%$ return reprices to the base price with a P&L of exactly $\$0.000000$. P&L is monotone in the underlying move across all 250 scenarios, as a long put requires. And the P&L is convex: symmetric up and down moves of 0.5% , 1% , 2% and 3% sum to $+0.1221$, $+0.4899$, $+1.9815$ and $+4.5188$ respectively, where a linear position would sum to exactly zero on every row. That convexity is the entire reason full revaluation is required, and Figure 6 shows what ignoring it would cost.

4.1.7 Value-at-Risk

Applying (17) and (18) to the 250 scenario P&Ls gives a 99% one-day VaR of **$\$2.4488$** and an expected shortfall of **$\$2.5805$** , or 36.89% and 38.87% of the $\$6.6391$ position value. The arithmetic, reproduced in full in Table 5, is deliberately transparent: the fractional rank $h = (250 - 1) \times 0.01 =$

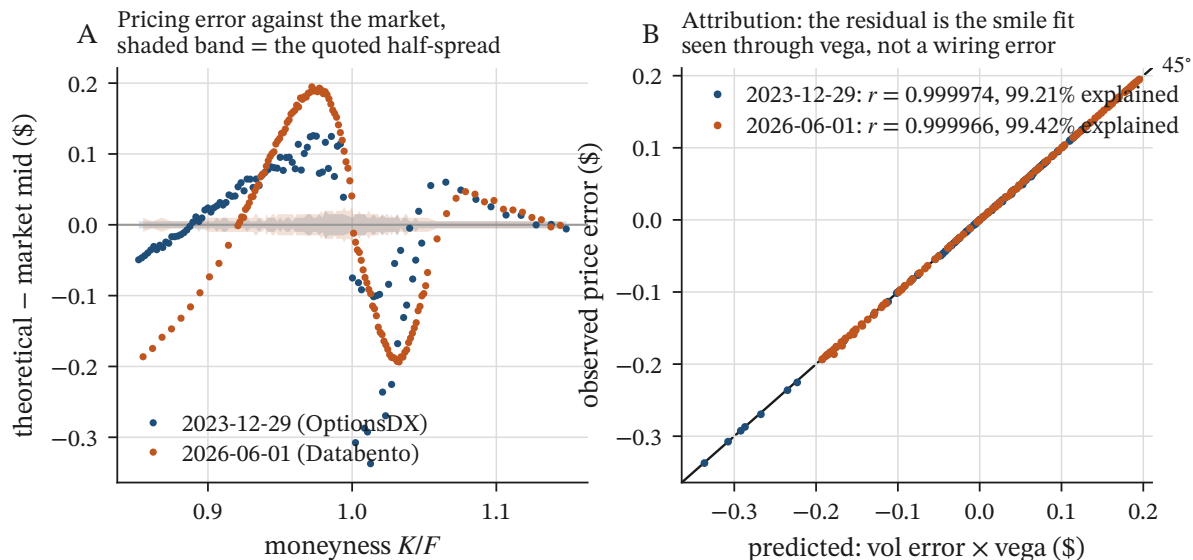


Figure 5: **A:** theoretical price minus market mid across the calibration band, with the quoted half-spread shaded. The errors are an order of magnitude larger than the spread — SPY spreads here have a median of \$0.010 — so “inside the spread” is not an achievable standard for a smile model and absolute error is the meaningful measure. The oscillating shape is the SABR residual of Figure 2 seen through vega. **B:** the attribution test. Each strike’s observed pricing error is plotted against the error predicted from its volatility fit error multiplied by its vega. The points lie on the 45° line: 99.21% and 99.42% of the pricing error is explained in the two cases, with correlations of 0.999974 and 0.999966. A wiring error — spot passed where the forward belongs, a mismatched day count, a wrong carry — would be structural and would not track the volatility fit. It does track it, so what remains is the smile fit, which is limited by the data rather than by the plumbing.

2.49 falls between the third and fourth worst scenarios, and the quantile is their weighted blend.

An independent recomputation with `numpy.percentile` using the linear method agrees to 0.000×10^0 — it returns -2.448845174615 against our -2.448845174615 . The agreement is exact because NumPy’s default uses the same $(N - 1)p$ rank definition we implemented, which is precisely why it is a good check on the arithmetic and a poor substitute for the implementation: calling it would have meant adopting whichever convention the library happens to default to rather than stating one.

4.2 Case 2: 2026-06-01, Databento

The same code was run against the Databento-sourced dataset with only the configuration changed. Every stage behaved, and in two respects behaved better than on the original data.

All 197 out-of-the-money strikes inverted successfully, none failed. The SABR fit is *better* than on the original chain, with an RMSE of 0.2080 vol pts over 125 strikes against 0.2346 over 103 — evidence that the pipeline is not overfitted to the inputs it was developed on. The attribution test of Section 4.1.5 gives 99.42% explained with a correlation of 0.999966, essentially reproducing the first case’s result on independent data.

Three features of the market state differ materially and are worth stating before the risk numbers, because they are what the risk numbers reflect.

The carry regime flips. The implied dividend yield is +1.8999% against -0.1926% in 2023.

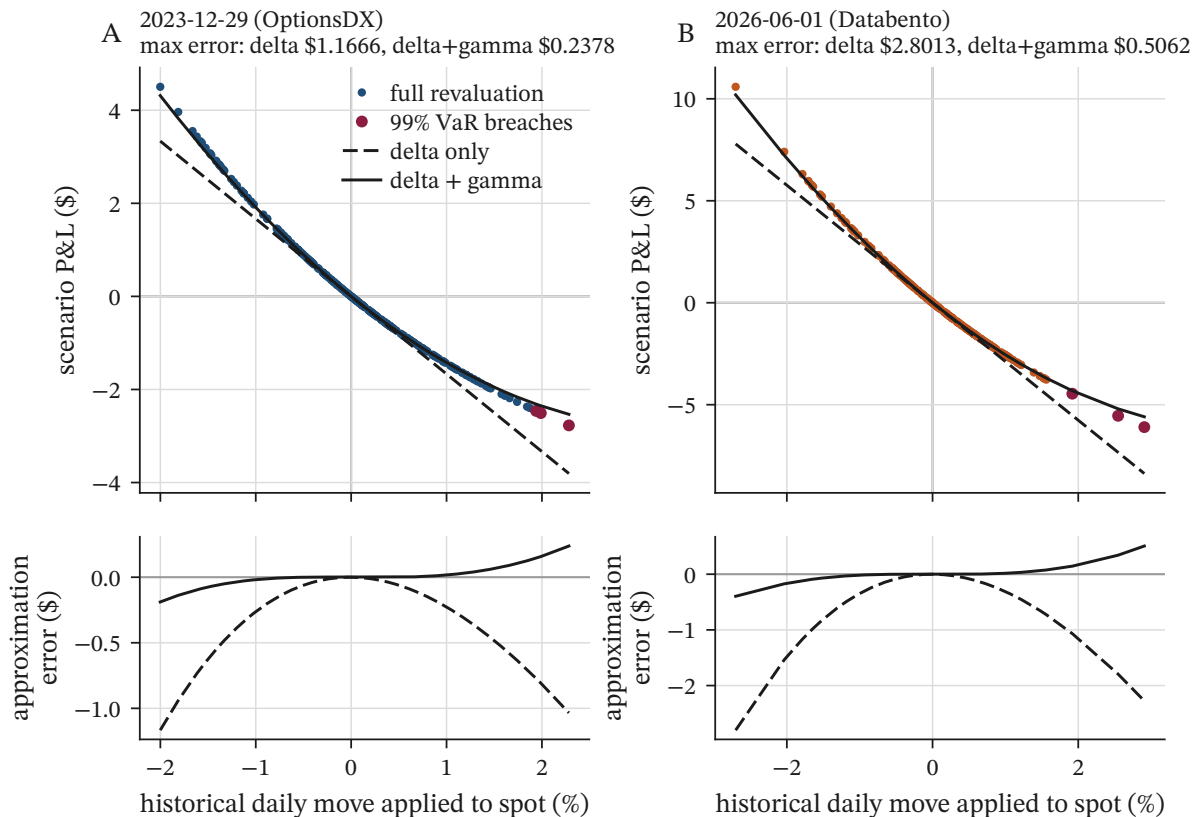


Figure 6: The filing’s prohibition, priced. Points are fully revalued scenario P&Ls; the lines are the delta-only and delta-plus-gamma expansions of (16), computed with Greeks taken from the full chain including the smile’s response to spot — that is, the most favourable version of the shortcut available. The lower panels show the approximation error directly. Delta alone is wrong by up to \$1.17 in the 2023 case, 17.6% of the whole \$6.64 position value. Delta plus gamma is much better but fails where it matters: on the single worst 2023 scenario it returns $-\$2.534$ against a true $-\$2.772$, *understating* the worst-case loss by 8.6%. Note also that every VaR breach (wine) lies on the *right*: a long put loses when the market rallies, so the scenarios driving this VaR are the best days of the year, not the worst.

This is a calendar fact rather than a modelling artefact: one SPY ex-dividend date falls between 2026-06-01 and 2026-07-17 where none fell in the 2023 window. Stated in dollars rather than annualised, $+1.8999\%$ is \$1.82 of dividend on a \$758.54 underlying, which is SPY’s actual quarterly payout. Nothing in the calibration forces this; it falls out of put-call parity applied to the quotes. The consequence runs through the pricing: with $b < r$ an American call now *does* carry early-exercise value, and the put’s premium is larger here than in the first case, so the choice of a put as the reference position holds for the opposite reason.

Rates fell, 5.40% to 3.66%, pinned externally from the same FRED series in both cases.

The skew steepened and the volatility level rose — $\hat{\rho}$ from -0.524 to -0.634 , at-the-money implied volatility from 11.76% to 13.94% — against *lower* realised volatility in the lookback window, 13.14% to 11.88%. A higher implied volatility on a calmer realised window is a genuine feature of the 2026 surface.

The resulting 99% one-day VaR is **\$4.1216** with an expected shortfall of **\$5.3674**, or 28.58% and 37.22% of the \$14.4221 position value. The full arithmetic is in Table 5 and the distribution in

Table 5: Value-at-Risk and expected shortfall for both cases, with the interpolation of (17)–(18) reproduced in full so that each figure can be rebuilt by hand. In both cases the fractional rank $h = 2.49$ falls between the third and fourth worst scenarios and the quantile is their weighted blend. Note that every bracketing scenario is an *up* day: a long put loses when the market rallies. The stress rows apply the same position and the same calibrated surface to a harsher return window, changing only the return distribution; they should be read as upper bounds for the reason given in Section 6.

	2023-12-29		2026-06-01	
	Baseline	Stress (2020)	Baseline	Stress (to 2026-01-02)
Scenarios N	250	250	250	250
Tail probability p	0.01	0.01	0.01	0.01
Fractional rank $h = (N - 1)p$	2.49	2.49	2.49	2.49
Lower order statistic $L_{(3)}$ (\$)	−2.462080	−5.052603	−4.459823	−5.631089
Upper order statistic $L_{(4)}$ (\$)	−2.435070	−4.806637	−3.769542	−4.797515
Interpolation weight w	0.49	0.49	0.49	0.49
Interpolated quantile $\hat{Q}$ (\$)	−2.448845	−4.932080	−4.121585	−5.222638
99% one-day VaR (\$)	2.4488	4.9321	4.1216	5.2226
as % of position value	36.89	74.29	28.58	36.21
Expected shortfall (\$)	2.5805	5.3881	5.3674	8.0276
breaching scenarios	3	3	3	3
Worst scenario P&L (\$)	−2.7719	−5.5885	−6.0986	−11.7889
produced by	2023-01-06	2020-03-13	2026-03-31	2025-04-09
that day’s SPY move	+2.283%	+9.290%	+2.907%	+10.502%
Stress / baseline	—	2.01×	—	1.27×
Position value V_0 (\$)	6.6391	6.6391	14.4221	14.4221

Figure 7.

4.3 The two cases compared

Table 6 sets the two runs side by side. They disagree substantially, and the question worth answering is whether the disagreement is the market or the plumbing.

It is the market. Set against differences of 68% in dollar VaR and 8.3 percentage points in VaR-to-position, a forward that agrees between vendors to 0.89 basis points and implied volatilities whose median difference is exactly zero cannot be carrying the result. The sharpest single regime difference is the shape of the tails: the 2023 window has excess kurtosis of -0.178 , meaning tails *thinner* than a normal distribution, while the 2026 window has $+1.569$. Both windows are calm by their own volatility measure; the second is calm in a fat-tailed way.

One asymmetry should be stated rather than glossed. The stress ratios, $2.01\times$ and $1.27\times$, are not comparable to each other. The first case’s dataset reaches 2020; the Databento pull spans seventeen months, so the harshest window available to the second case is the one containing the April 2025 selloff. That is a smaller multiple because April 2025 was a smaller event than March 2020, not because the method behaved differently.

4.4 The lookback window dominates the answer

The largest sensitivity in this study is not a modelling choice. It is which twelve months precede the reference date.

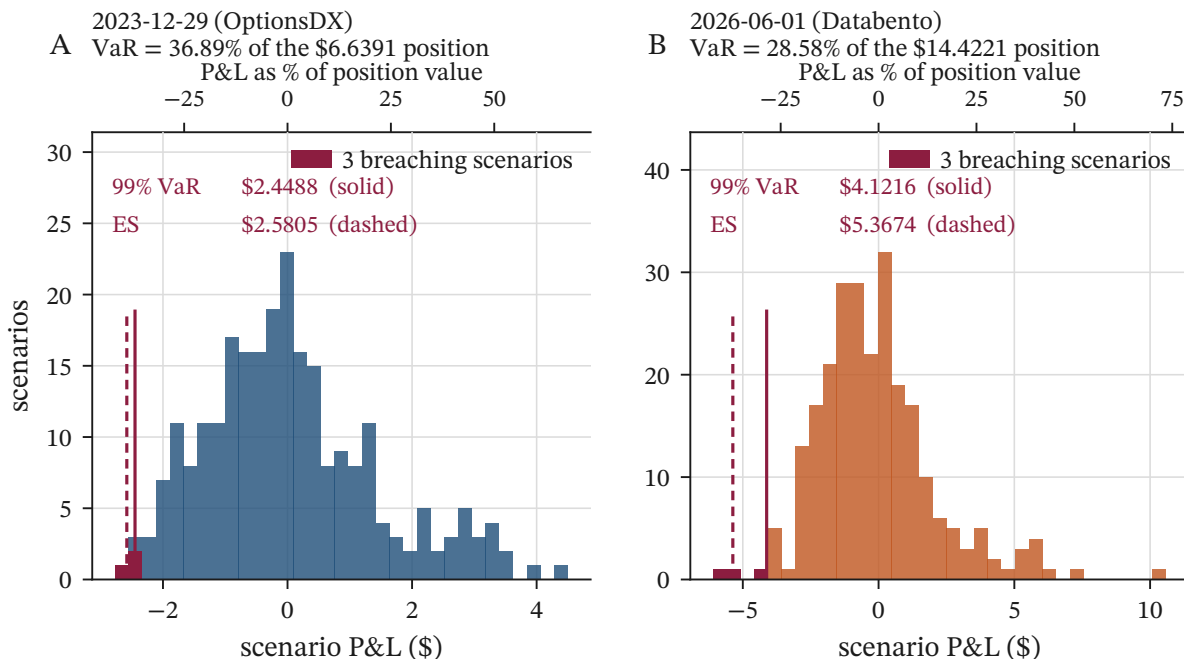


Figure 7: The fully revalued scenario P&L distributions, with the interpolated 99% threshold (solid) and expected shortfall (dashed) marked and the three breaching scenarios in wine. The secondary axis expresses P&L as a percentage of position value, which is the only like-for-like comparison between the two cases: the dollar figures are not comparable because the positions differ in strike and in underlying level. On that basis the 2026 case is the *less* risky of the two, 28.58% against 36.89%. Both distributions are right-skewed, which is gamma: a long option gains more from a large move than it loses from a move of the same size in the other direction. The 2026 distribution has the visibly longer right tail, consistent with its window's excess kurtosis of +1.569 against -0.178 .

Evaluating the same position by the same method as if standing at the end of each of several past years gives the upper block of Table 7. The 2023 window is the calmest of six by a wide margin and the only one with negative excess kurtosis; every other window shows the textbook fat tails. Its first percentile return, which is essentially what a 99% VaR reads, is -1.64% against -6.76% for a window ending in 2020 — a factor of 4.1. That factor is neither a modelling choice nor a market view.

The lower block of Table 7 makes the same point more directly, because it holds the dataset, the vendor and the underlying fixed and moves only the reference date. The April 2025 tariff selloff (-5.85% , followed five sessions later by $+10.50\%$) sits at sessions 63–68 of the Databento pull. It is inside the 250-day window for reference dates up to roughly 2026-03 and has aged out by 2026-04-28. Across that boundary the window's first percentile halves, from -3.56% to -1.75% , its annualised volatility falls from 18.88% to 12.51% and its excess kurtosis collapses from $+26.96$ to $+2.07$. Nothing about the market changed at that moment. Only the window did. Figure 8 shows both effects.

Table 6: The two reference cases side by side. The dollar VaRs are not directly comparable — different strikes on a different underlying level, and marginally different moneyness sign — so VaR as a percentage of position value is the like-for-like figure, and on that basis the second case is the less risky of the two. The upper block lists differences attributable to the market; the lower block bounds what the change of data source could contribute, measured on a shared date by the cross-vendor check of Section 3.3.

	2023-12-29 (OptionsDX)	2026-06-01 (Databento)
<i>Market state</i>		
Risk-free rate r	5.4000%	3.6600%
Cost of carry b	5.5926%	1.7601%
Implied dividend yield q	−0.1926%	+1.8999%
Carry regime	$b > r$	$b < r$
SABR $\hat{\rho}$ (skew)	−0.524225	−0.633712
At-the-money implied volatility	11.76%	13.94%
Fit RMSE	0.2346 vol pts	0.2080 vol pts
<i>Lookback window</i>		
Realised volatility (annualised)	13.14%	11.88%
Skewness	−0.028	−0.244
Excess kurtosis	−0.178	+1.569
Worst / best day	−2.00% / +2.28%	−2.70% / +2.91%
<i>Result</i>		
Position value (\$)	6.6391	14.4221
99% one-day VaR (\$)	2.4488	4.1216
VaR as % of position	36.89%	28.58%
Expected shortfall (\$)	2.5805	5.3674
Stress / baseline	2.01×	1.27×
<i>Data-source contribution, measured on the shared date 2023-12-29</i>		
Median mid price difference	\$0.0000 over 151 strikes	
Median re-implied IV difference	0.0000 vol pts	
95th percentile IV difference	0.2286 vol pts	
Parity forward difference	−\$0.0425 (0.89 bp)	
Re-implied crossover gap	reproduced to 0.0076 vol pts	

Table 7: Return-window statistics. **Upper block:** the 2023 case’s window against five earlier year-end windows from the same dataset. Ours is the calmest of the six and the only one with negative excess kurtosis — not one day in calendar 2023 moved beyond three standard deviations, against the 0.67 such days a fitted normal predicts, so the tails are *thinner* than normal rather than fatter. (These comparison years are context only and are not put through the calendar validation of Section 3; vendor data quality varies by year, and the reference window itself is clean.) **Lower block:** the 2026 case’s window against four earlier reference dates from within the same continuous seventeen-month dataset. The first percentile holds at -3.56% for three successive dates and then halves to -1.75% , because the April 2025 selloff leaves the 250-day window between them.

As of	Ann. vol	Skewness	Excess kurtosis	Worst day	1st percentile
<i>OptionsDX dataset, year-end windows</i>					
2011-12-30	23.08%	−0.433	+2.581	−6.51%	−4.34%
2015-12-31	15.65%	−0.231	+2.117	−4.13%	−2.78%
2018-12-31	17.18%	−0.409	+3.148	−4.21%	−3.21%
2020-12-31	34.14%	−0.598	+7.631	−11.51%	−6.76%
2022-12-30	24.09%	+0.026	+0.362	−4.34%	−3.74%
2023-12-29 (Case 1)	13.14%	−0.028	−0.178	−2.00%	−1.64%
<i>Databento dataset, rolling reference dates</i>					
2026-01-02	19.48%	+1.514	+23.636	−5.85%	−3.56%
2026-01-30	19.30%	+1.555	+24.644	−5.85%	−3.56%
2026-03-16	18.88%	+1.703	+26.960	−5.85%	−3.56%
2026-04-28	12.51%	+0.068	+2.067	−2.70%	−1.75%
2026-06-01 (Case 2)	11.88%	−0.244	+1.569	−2.70%	−1.75%

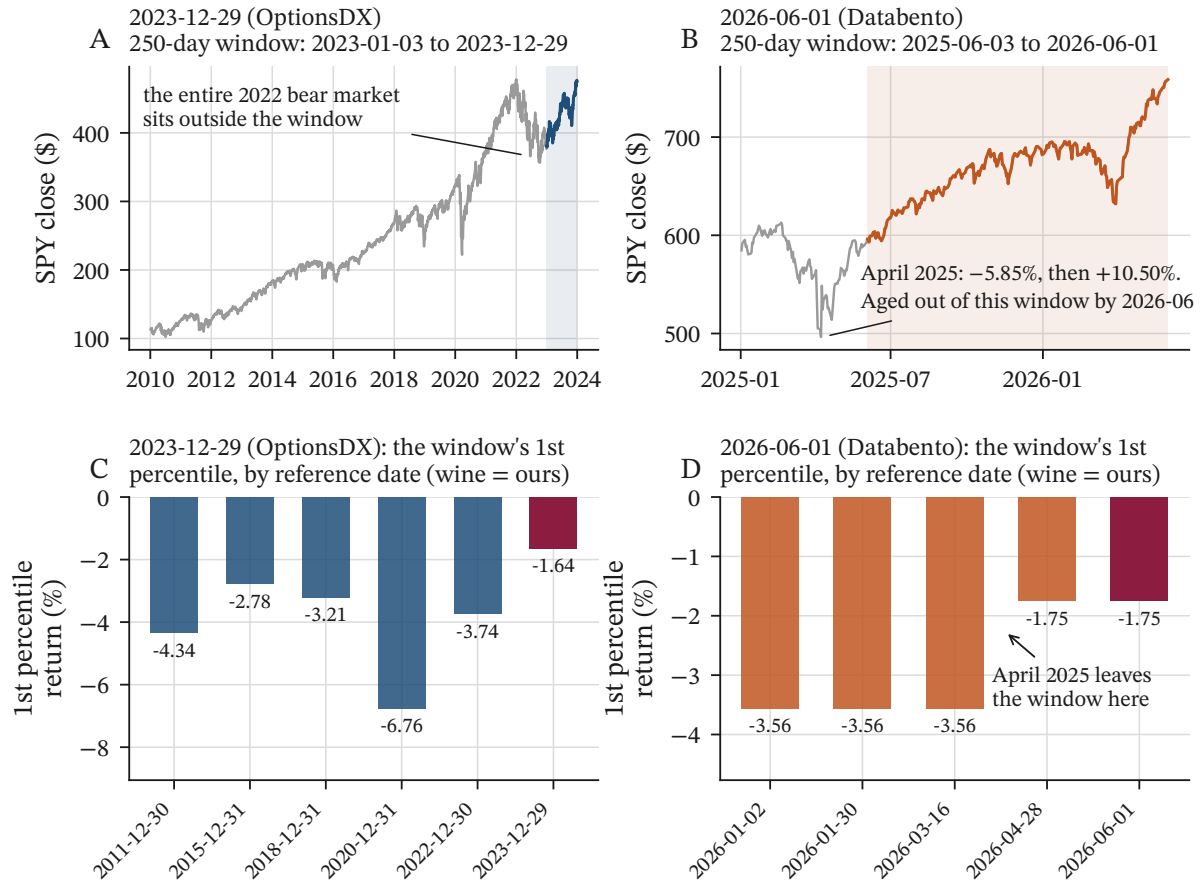


Figure 8: The dominant sensitivity in the study. **A, B:** the SPY price path for each dataset with the 250-day lookback window shaded. In the 2023 case the entire 2022 bear market — which took SPY from \$477 to \$357 — sits outside the window and contributes nothing to the VaR. In the 2026 case the April 2025 selloff sits just outside the shaded region for the same reason. **C, D:** the window's first-percentile return as the reference date moves, same position and same method throughout. Panel D is the sharper demonstration because it holds the dataset, the vendor and the underlying fixed: the first percentile is unchanged at -3.56% for three successive reference dates and then halves to -1.75% , purely because April 2025 has fallen off the back of the window. This is the central weakness of unweighted historical simulation — a crisis counts fully until the day it leaves the window, and not at all afterwards.

5 Robustness and diagnostics

5.1 The put/call inconsistency: what was an artefact and what was not

Section 4.1.3 showed that re-impling the volatilities removed a wrong-signed kink at the put/call crossover and improved the fit. It did not remove the inconsistency entirely, and the cross-vendor data of Section 3.3 allows the remainder to be separated into a vendor artefact and a market feature.

The kink was the vendor's. The OptionsDX volatilities stepped +0.291 vol pts upward across a downward-sloping smile. Databento does not reproduce this. Relatedly, the two crossed put quotes at $K = 477$ and $K = 478$ (bid 7.36 / ask 7.33, and bid 7.78 / ask 7.77) that forced those strikes out of the OptionsDX smile are clean and two-sided in the Databento data (7.37 / 7.40 and 7.80 / 7.83), which contains zero crossed quotes in the entire out-of-the-money chain. Both were vendor defects. The second reference case corroborates this independently: in 2026 both the pre-correction and post-correction gaps carry the correct negative sign (−0.1227 and −0.0436), so an internally consistent starting point does not produce the kink.

The residual is real. Measuring the crossover anomaly as the adjacent gap minus what the put wing's own slope predicts, using a slope fitted over a range of strikes rather than a two-point difference:

	Adjacent gap	Fitted wing slope	Predicted gap	Residual
OptionsDX, 476 → 479 (\$3 span)	−0.2088	−0.1264/\$1	−0.3792	+0.1704
Databento, 476 → 479 (\$3 span)	−0.2164	−0.1287/\$1	−0.3860	+0.1697
Databento, 478 → 479 (\$1 span)	+0.0208	−0.1254/\$1	−0.1254	+0.1462

All quantities in volatility points. The two vendors agree on identical strikes to 0.0007 vol pts, and closing the \$3 gap left by the crossed OptionsDX quotes — possible only in the Databento data — moves the answer by 0.024. **The residual put/call inconsistency is a genuine market feature of roughly +0.15 vol pts, not a vendor artefact.**

We tested two explanations and neither closes it. It is not the forward: the gap would vanish at $F \approx \$479.6$, some \$0.70 above the parity forward and 3.5 standard deviations of that estimate, so not noise. Correcting the known European-for-American bias in (1) — subtracting each leg's early-exercise premium before applying parity — moves the forward from \$478.892 to \$479.036, which is +\$0.144, about 20% of the gap. The bias is real and correctly signed but it is not the main cause. The most likely remainder is that the parity forward is computed from both legs, one of which is always the illiquid in-the-money one. We report this as unresolved.

A diagnostic of ours that the second source falsified. This is worth recording because it is a negative result about our own instrument, not about the data. Our crossover diagnostic originally predicted the expected gap from a *two-point* local slope — the last two put quotes only — a choice made deliberately to keep the measure local. On this date that choice fails:

	Two-point slope	Predicted gap	Residual
OptionsDX, 476 → 479	−0.0193/\$1	−0.0578	− 0.1510
Databento, 476 → 479	−0.1084/\$1	−0.3253	+ 0.1089
Databento, 478 → 479	−0.1126/\$1	−0.1126	+0.1334

Table 8: Sensitivity of the reported VaR to the quantile rank convention of (17). All three are in common use and the filing specifies none of them (Hyndman and Fan, 1996). In the first case the ambiguity is worth 1.46% of the VaR, small enough to be dismissed. In the second it is worth 21.26%, because the Weibull rank of 1.51 lands between the second and third worst scenarios of a fat-tailed sample where the first case’s thin-tailed sample is nearly flat. The wording’s indeterminacy is therefore not a fixed small quantity but one that scales with the shape of the tail — which is precisely the regime in which a margin figure matters most.

Convention	2023-12-29		2026-06-01	
	Rank h	VaR (\$)	Rank h	VaR (\$)
$(N - 1)p$ (R type 7, NumPy default, ours)	2.4900	2.4488	2.4900	4.1216
Np	2.5000	2.4486	2.5000	4.1147
$(N + 1)p - 1$ (R type 6, Weibull)	1.5100	2.4844	1.5100	4.9909
Spread across conventions (\$)	0.0358		0.8762	
as % of the reported VaR	1.46%		21.26%	

The residual *changes sign between vendors on the same strikes*. The cause is a single quote: the 475 $\rightarrow$ 476 put volatility step is -0.0193 vol pts in the OptionsDX data against -0.1084 vol pts in Databento, because the OptionsDX \$475 put mid is seven cents low. A two-point slope has no averaging, so a one-cent quote error propagates undamped — here it flattened the slope by a factor of 5.6 and flipped the sign of the conclusion. The fitted-slope measure is stable across vendors ($+0.1704$ against $+0.1697$) and across strike pairs, and is the measure the paragraph above uses. Any figure computed from the two-point version, including one we previously recorded, should not be relied upon.

5.2 The rank convention is a larger ambiguity than the first case suggests

The filing’s phrase “linear interpolation to the 99% threshold” does not determine a calculation, because it leaves open where in index space the threshold sits. Table 8 recomputes both cases’ VaR under the three conventions in common use.

Reporting only the first case would have supported the comfortable conclusion that the convention barely matters. It does not generalise. At $N = 250$ the three ranks differ by at most one position in the order statistics, so the spread is governed entirely by how steep the P&L distribution is between the second and fourth worst scenarios — nearly flat in the thin-tailed 2023 sample, steep in the fat-tailed 2026 one. The ambiguity also grows as the scenario count falls. A specification that intends a reproducible number must name the convention.

5.3 The cost of the approximations, collected

Three sources of error have been measured rather than assumed, and it is worth setting them beside one another because they are of very different sizes.

The BAW approximation contributes approximately 1% to every scenario P&L (Section 4.1.4), amplified $5.5\times$ from the price level because its sign changes with moneyness. It is not removable within the specification, since the filing names BAW; a converged lattice would remove it at roughly a thousandfold increase in runtime.

The rank convention contributes 1.46% or 21.26% depending on the case (Section 5.2). It is removable at zero cost by naming a convention.

The data source contributes a median of exactly zero (Section 3.3), with a 95th-percentile implied-volatility difference of 0.2286 vol pts and a forward agreeing to 0.89 basis points.

Against these, the lookback window moves the first-percentile return by a factor of 4.1 across the windows in Table 7, and by a factor of 2.0 within a single continuous dataset over four months. The ordering is not close.

6 Limitations

The following bound what these results establish. Several are consequences of matching the filing rather than defects, and are marked as such.

Neither lookback window contains a crisis. This is the dominant limitation and it is quantified in Section 4.4. Calendar 2023 had realised volatility of 13.14%, excess kurtosis of -0.178 and not one day beyond three standard deviations; its first percentile return is -1.64% against -6.76% for a window ending in 2020. Both reported VaRs are benign as a consequence of their windows, not of their positions. The unweighted window is specified by the filing, so this is intended behaviour, but it means neither number should be read as an estimate of what the position could lose in a stressed market.

There is no independent volatility risk in either number. The surface is frozen at its reference calibration and only the anchor point moves with spot, so scenario P&L contains spot risk and the smile's response to spot and nothing else. For a long option this matters a great deal. On the worst 2020 stress scenario — a $+9.29\%$ day — the frozen surface gives a loss of \$5.59; lifting volatility by ten points cuts that to \$1.90 and by twenty points turns it into a \$3.32 gain. A genuine joint replay would shock the surface with the same historical day's surface, and the historical surfaces to build that do exist in the dataset. It was not built.

The stress figures are therefore upper bounds, in a direction that must be stated explicitly. A long put is long vega and loses on rallies, so holding today's calm surface instead of the stressed period's elevated one makes the losing tail *deeper* than a true joint simulation would. The \$4.9321 stress VaR overstates what a full joint replay of 2020 would give.

Sticky-moneyness is the less conservative convention for these positions. Holding the SABR parameters fixed and re-evaluating at the new forward pins the smile to moneyness. Because the equity smile slopes downward in strike, a rally lowers K/F and slides a fixed strike into the steeper part of the wing, so the reference put's volatility *rises when the market rises* — which reads backwards against the intuition that volatility spikes in a selloff, but that intuition concerns the level of the whole surface, which this model deliberately freezes. Against a sticky-strike alternative the convention dampens P&L in both directions, and since a long put loses on rallies, it reduces the reported VaR.

The two stress windows are not comparable to each other. March 2020 and April 2025 are events of different magnitude. The $2.01\times$ and $1.27\times$ ratios describe each case's own dataset and should not be read against one another.

The 2026 case has no external price benchmark of its own. The 2023-12-29 state can be checked against two independent sources. The 2026-06-01 state rests on one, cross-validated only in the sense that the *source* was verified on a different date. A second feed for 2026-06-01 would cost cents and has not been purchased. This is a weaker claim than the first case can make and should be read as such.

One position, one day, one underlying. Two single options, not portfolios. No multi-day

scaling, no correlation across underlyings, no diversification analysis. A portfolio extension — several strikes across at least two expiries, with each expiry given its own SABR calibration, since reusing one calibration across tenors is a common and expensive error — was scoped and deliberately not built.

Expected shortfall averages three scenarios. At $N = 250$ and $p = 0.01$ only three P&Ls breach the threshold, so the reported ES is correspondingly noisy. This is a limitation of the scenario count, not of the estimator. One route worth exploring, and not taken here, is to fit the tail rather than average it: a Generalized Pareto Distribution fitted to exceedances over a lower threshold, in the extreme-value-theory tradition, would draw on many more than three observations and would deliver a smoother expected shortfall *without* lengthening the 250-day window the filing specifies. Whether the resulting stability justifies introducing a distributional assumption into a method that is otherwise wholly non-parametric is the question such an extension would have to settle.

End-of-day marks only, at theoretical mid. One snapshot per session in both cases. An end-of-day mark says nothing about the intraday path a margin model would care about, and positions are marked at theoretical mid with no liquidity or bid-ask adjustment — for a large position in the far wings, the cost of actually exiting is not the mid.

Two measurement caveats specific to this build. The Databento pre-close window is a three-minute consolidated best-bid/offer rather than a true end-of-day snapshot, and the rule for reducing it to one is ours (Section 3.2). And the volatility columns in the 2026 dataset are this project’s own European inversions, so the before/after comparison in Figure 3A measures the European-to-American correction alone in that case, rather than correction plus vendor error as in 2023.

7 Conclusion

Implemented on SPY options, the methodology specified in SR-FICC-2013-02 does what it sets out to do. Its three components fit together as intended: the SABR curve supplies a volatility at any strike, the quadratic approximation converts it to an American price, and full revaluation propagates both through 250 historical scenarios without recourse to a Taylor expansion. The joined models price the market with a mean signed error of $-\$0.0007$ across the calibration band, and 99.21% of the residual is attributable to the volatility fit seen through vega rather than to the plumbing.

Two aspects of the specification are less determinate than the wording suggests, and both are worth naming for anyone implementing it. The filing names SABR but not β , and β is close to jointly unidentifiable with ρ : across its entire admissible range the fit error moves by 0.04 vol pts while $\hat{\alpha}$ swings by a factor of 476. And the filing calls for linear interpolation to the 99% threshold without naming a rank convention, an ambiguity worth 1.46% of VaR in one of our cases and 21.26% in the other. Neither is fatal; both should be pinned down explicitly rather than inherited from whichever library is at hand.

The methodology also carries a precision floor it cannot escape. BAW’s error changes sign with moneyness, so it does not cancel when prices are differenced into scenario P&Ls but amplifies 5.5 $\times$, contributing approximately 1% to every reported figure. The filing specifies BAW, so this is intended behaviour rather than a defect — but it is the resolution limit of any number produced this way.

None of these is the dominant sensitivity. That is the composition of the lookback window, and the second reference case demonstrates it more directly than a comparison across eras can: holding

the dataset, the vendor, the underlying and the method fixed and moving only the reference date, the window's first-percentile return halves from -3.56% to -1.75% as the April 2025 selloff falls off the back of a rolling 250-day window. Nothing about the market changed at that moment. A margin figure computed this way is, to first order, a statement about the preceding twelve months.

Finally, the cross-vendor validation is the part of this study we would repeat first in any similar exercise. It cost four cents. It established that a wrong-signed kink which had motivated a substantive methodological choice was a vendor artefact; it confirmed that the residual inconsistency beneath that artefact was real, to 0.0007 vol pts across two independent sources; it falsified a diagnostic of our own whose sign depended on a single seven-cent quote; and it is the sole reason the difference between our two reference cases can be attributed to the market with a bound rather than an assertion.

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